MATRIX · TRACE

Trace properties at a glance

Seven identities, each linked to the section that proves it. What holds them together is that the trace sees only the diagonal — every entry below either exploits that or is limited by it.

07
PROPERTIES

UNCONDITIONAL 3

i Linearity $\operatorname{tr}(cA + dB) = c\operatorname{tr}(A) + d\operatorname{tr}(B)$

UNCONDITIONAL

REQUIRES A, B both $n \times n$

The trace is a linear functional on the space of $n \times n$ matrices — it maps matrices to scalars while respecting addition and scaling. That is a stronger statement than the formula alone: it means the trace lives in the dual space, which is what makes the Frobenius inner product possible.

ii Transpose invariance $\operatorname{tr}(A^{\mathsf{T}}) = \operatorname{tr}(A)$

UNCONDITIONAL

REQUIRES A square

Transposition reflects entries across the main diagonal, and the diagonal entries are exactly the ones fixed by that reflection. Since the trace reads only those, it cannot notice.

vi Commutator trace $\operatorname{tr}(AB - BA) = 0$

UNCONDITIONAL

REQUIRES A, B both $n \times n$

Immediate from the cyclic property and linearity. The consequence is sharper than it looks: since $\operatorname{tr}(I_n) = n \neq 0$ over $\mathbb{R}$ or $\mathbb{C}$, the identity matrix is never a commutator. No pair of matrices satisfies $AB - BA = I$ in finite dimensions — which is exactly why the canonical commutation relation of quantum mechanics needs infinite-dimensional operators.

CONDITIONAL 4

iii Cyclic property $\operatorname{tr}(AB) = \operatorname{tr}(BA)$

CONDITIONAL

HOLDS WHEN both products defined — A and B need not be square

COMMON ERROR reading it as full commutativity: $\operatorname{tr}(ABC) = \operatorname{tr}(BCA)$ but not $\operatorname{tr}(BAC)$

Rotating the factors is allowed; permuting them arbitrarily is not. For three matrices the cyclic rotations $ABC \to BCA \to CAB$ all share a trace, while BAC generally does not. This is the identity everything below is derived from — it is the reason the trace survives a change of basis at all.

> **WHERE IT BREAKS**
>
> `A = [[1, 0], [0, 0]], B = [[0, 1], [0, 0]], C = [[0, 0], [1, 0]]`
> `tr(ABC) = 1 but tr(BAC) = 0`

iv Similarity invariance $\operatorname{tr}(P^{-1}AP) = \operatorname{tr}(A)$

CONDITIONAL

HOLDS WHEN P invertible

A direct consequence of the cyclic property: rotate P^{-1} to the back and it meets P. The consequence is what matters — the trace is a property of the underlying linear transformation, not of the basis chosen to write it down. Two matrices with different traces cannot represent the same map.

v Eigenvalue sum $\operatorname{tr}(A) = \lambda_1 + \lambda_2 + \cdots + \lambda_n$

CONDITIONAL

HOLDS WHEN eigenvalues counted with algebraic multiplicity, over $\mathbb{C}$

Spectral information read straight off the diagonal, with no characteristic polynomial to solve. Over $\mathbb{R}$ the statement can fail because some eigenvalues may not exist there — a rotation matrix with $\theta \neq 0, \pi$ has trace $2\cos\theta$ and no real eigenvalues at all. The sum is still correct once the complex pair is counted.

vii Frobenius inner product $\langle A, B\rangle_F = \operatorname{tr}(A^{\mathsf{T}}B)$

CONDITIONAL

HOLDS WHEN real entries; use A^* for the complex case

Gives the space of matrices a geometry — angles, lengths and projections, exactly as the dot product does for vectors. Positive definiteness holds because $\langle A, A\rangle_F$ is the sum of every entry squared, so the induced norm is the Frobenius norm.

Two of these are load-bearing for the rest. Linearity makes the trace a functional rather than merely a formula, and the cyclic property is what similarity invariance, the eigenvalue sum and the commutator identity are all derived from.